

Alexander King, PhD

Applied AI Researcher & Computational Scientist

Remote (based in Riverside, CA)

alexander.julian.king@gmail.com

(925) 451-6310

alexanderjulianking.github.io

linkedin.com/in/alexander-julian-king

github.com/AlexanderJulianKing

PROFILE

Applied AI researcher and computational scientist who builds and evaluates systems for problems where plausible outputs are not enough. Neuroscience PhD and first-author in the *Journal of Neuroscience*; independent work spans agent failure analysis, LLM benchmarks, released scientific-audit software, RAG, and source-grounded automation. Works end to end from question framing through controlled evaluation and evidence-linked reporting.

AI EVALUATION & RELIABILITY

Silent Precondition Traps for Data-Science Agents

2026

Independent research study, under review at TMLR (2026)

[Zenodo](#) | [GitHub](#)

- Identified a repeatable class of agent failures: familiar tools silently violate an uninspected data precondition, producing confident wrong answers. Two frozen 10-task batteries, 10 model-scaffold configurations, 1,000 runs: **26 to 100% failure rates**; 329 of 330 answered-but-wrong runs returned the exact planted error.

SC-Referee

2026

Selected for Anthropic's Built with Claude: Life Sciences hackathon (Builder track)

[GitHub](#)

- Released Python application that reconstructs analysis designs from heterogeneous data, results, and code, then runs deterministic statistical audits with abstention-aware, evidence-linked verdicts. On a published melanoma dataset, **16,289 "significant" genes fell to 770 (95.3%)** when rebuilt at the biological-sample level.

APPLIED AI SYSTEMS

Project Soothsayer

Solo-built

LLM evaluation & score-prediction system

[GitHub](#)

- Predicts Arena human-preference ELO from benchmark data alone (138 models, 122 features, ~40% missing) with custom imputation and adaptive local regression; $R^2 = 0.942$, **RMSE 13.90**, $\rho = 0.966$ (5-fold CV; **0.937** walk-forward).

Newscaster

Live since 2024

Automated daily AI-news podcast, 525+ episodes

[YouTube](#) | [GitHub](#)

- Automated daily pipeline with a multi-provider LLM router, LangGraph source-grounded research agent, hand-built cross-episode RAG memory (**recall@10 $\approx$ 0.91**), and self-correcting pre-TTS fact-checker.

COMPUTATIONAL BIOLOGY & GENOMICS

University of California, Riverside

2019–2026

Ph.D., Neuroscience (Computational Focus)

Riverside, CA

- First-author *Journal of Neuroscience* paper: discovered cell-type-specific alternative splicing of *Adgrl2* in single-cell RNA-seq (100K+ cells), then validated it at the bench with mouse genetics, blood-brain-barrier assays, and confocal imaging.
- Reproducible Python and R pipelines for single-cell and bulk RNA-seq on HPC/SLURM; integrated 6+ public GEO datasets. Trained 5 researchers; 4 years teaching, sole instructor for an upper-division course.

PUBLICATIONS

- King, A.**, et al. (2026). Endothelial *Adgrl2* expression and alternative splicing controls the cerebrovasculature. *Journal of Neuroscience*. (first author)
- Bern, M.^{*}, **King, A.**^{*}, Applewhite, D.A., & Ritz, A. (2019). Network-based prediction of polygenic disease genes involved in cell motility. *BMC Bioinformatics*. (* co-first)

SKILLS & EDUCATION

LANGUAGES & TOOLS Python, R, SQL, Bash; Git, pytest, HPC/SLURM, Google Cloud Platform, API integration

ML & STATS scikit-learn, pandas, NumPy; regression, imputation, PLS/PCA/SVD, conformal prediction

AI & EVALUATION agent failure analysis, benchmark design, LLM-as-judge, RAG/embeddings, LangGraph

BIOINFORMATICS single-cell/bulk RNA-seq, Seurat, Scanpy, differential expression, alternative splicing, GEO

EDUCATION Ph.D. Neuroscience, UC Riverside (2019–2026); B.S. Neuroscience, Reed College (2015–2019)